GE Credit Lending Model

Introduction

Business Understanding

The Business

- General Electric (GE) is an internation company
  - Aerospace
  - Energy
  - Healthcare
- Customers can open lines of credit to purchase good and services with GE's Credit department
- Customers receive differing amounts based on factors like needs, credit history, credit utilization, and accounts

The Problem

GE relies heavily on human intuition and complex mathematical formulas to judge the amount of credit to lend

Prone to error, imprecise, incorporates human bias

Lending errors put GE at risk for under allocation, causing customers to get the services they need elsewhere

Lending errors put GE at risk for over allocation, allowing customer to borrow more than they can pay off (default)

The Goal

Train an accurate linear regression model that is able to look at the customer data available to GE, and make predictions on how much credit to extent to a customer

Central Concerns

The model needs to be accurate or else it runs the same risks

- Model over allocation also introduces default risks
- Model under allocation also introduces customer loss risk

The model needs to be accurate in order to prevent the loss of money from the department. Need enough good quality data that can explain the factors the make up the target loan amount

Data Understanding

The Data

- 5,000 training observations and 100 validation observations
- 74 numerical features
- Feature categories:
  - Borrower information
  - Credit history
  - Credit utilization
  - Account counts

Target Variable Metrics

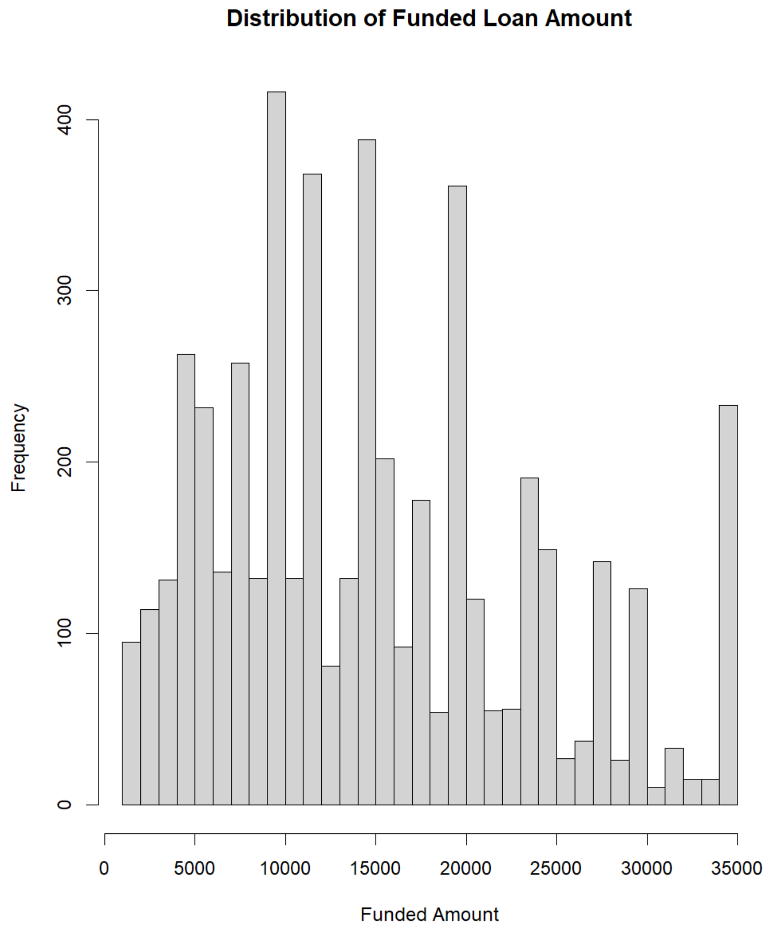
Target Variable: “Target Amount”

Minimum Loan: $1,000

Maximum Loan: $35,000

Average Loan: $15,238

Noticeable spikes around round numbers like 5k, 10k, 15k and 20k suggest human influence in the lending amount



Missing Values and Errors

Missing values made up less than 2% of the dataset

One error was found in the data where “-” was entered in a numerical column

Inclusion and Exclusion

In order to be included in the training dataset:

- Correlate with the target variable
- Not correlate with other predictors
- Not leak the loan amount

Top 5 Correlators

“Max Balance owed on all Revolving Accounts” (42%)

“Annual Income” (41%)

“Total Revolving Credit Balance”(33%)

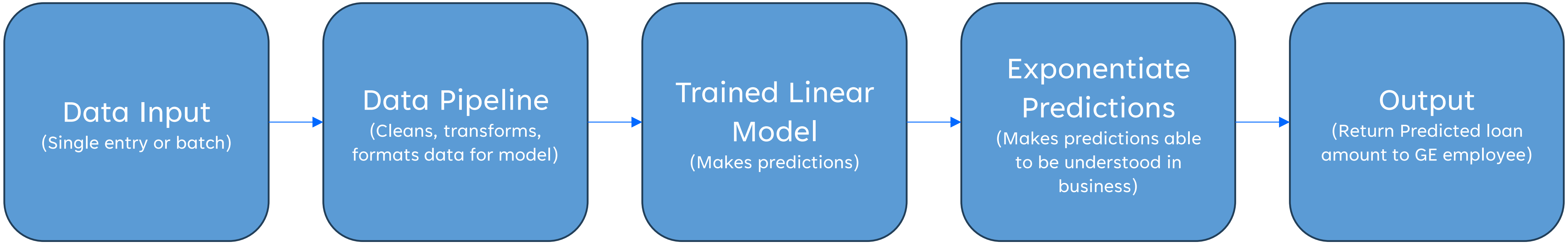
“Total Current Balance of all Accounts” (29%)

“Number of Satisfactory Bankcard Accounts” (23%)

Plan Definition

Data Preparation

Data Architecture Pattern



Data pipeline (Integrity)

Removed variables that were derived after the loan amount or uniquely identified the loan

Imputed (replaced) missing values with medians

Transformed all data logarithmically as it proved to make relationships to the target more linear, and reduced skew

Features that did not correlate to the loan among were dropped

Features that were highly correlated with each other were dropped

Error Handling (Quality)

1

The "-" error was handled by replacing the rouge value with NULL

2

Missing values were imputed (replaced) with the column median

Security Measures

Financial data is very sensitive, personal financial data is even more so

- The GE model is trained with no personally identifiable information.
  - No ethnicity or race
  - No age
  - No gender
  - No religion

Removes the chance for bias, or data leaking, and is incompliance with the ECOA (Equal Credit Opportunity Act)

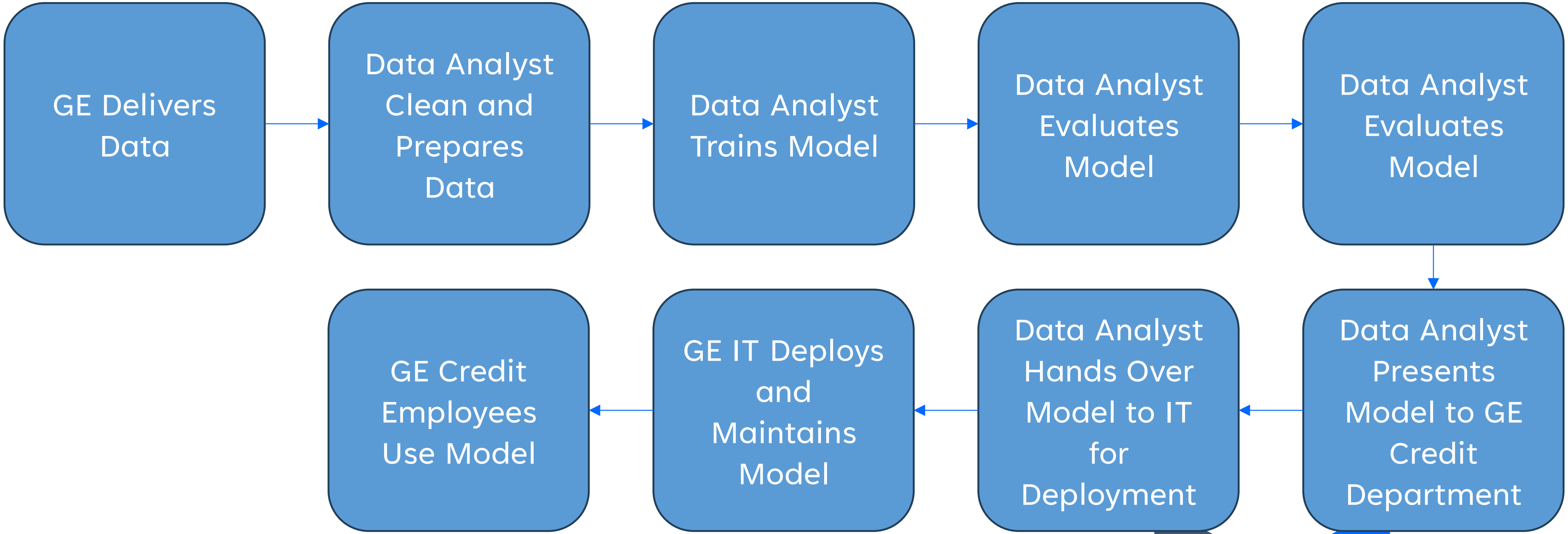
Stakeholders

GE Credit Employees: Give the model input and utilize model outputs

GE Credit Management: Oversee the department manage high level decisions

GE IT Employees: Manage the model quality (RSME, MAPE, and R2) and oversee deployment and maintenance

Project Timeline



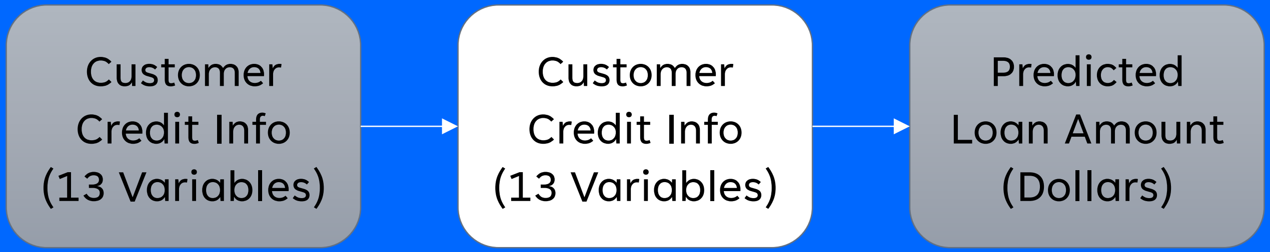
Professional and Effective Collaberation

- Regular Check ins with the Credit team to ensure that business need are correctly interpreted and being met
- R code is documented, commented, and reproducable
- Explain technical concepts in a way in which all audiences can understand
- Share interim results to flag concerns or change strategies

Plan Implantation

Deployment Plan

- This model, if deployed, will act as a decision support tool and not an automation to the process
- GE Credit employees will still be responsible for gathering and verifying customer information, and have the ultimate say in the end loan amount
- GE IT is responsible for hosting the model and data pipelines, ensuring it receives clean input data in the correct format



Plan Monitoring

When handing off the model, GE IT will be given the following metrics in the turnover report

RMSE: 7663.22 **MAPE: 55.73%** **R2: 0.25**

- Should be compared against the model's performance monthly on new test data with known results
- If the distribution of the trained features shifts, then the model may not be accurate either
- Monitoring the performance and metrics is crucial to maintaining the quality results

Plan Maintenance

Since all of the code is being handed over to GE IT, it is very simple to maintain the mode

- Simply change the source path of the data to a new recent snapshot of some customer data
- Re-run the script and train a new model on updated data
- Compare the results against each other to ensure it was a model weight's problem and not a deeper issue

Project Review

Project Review

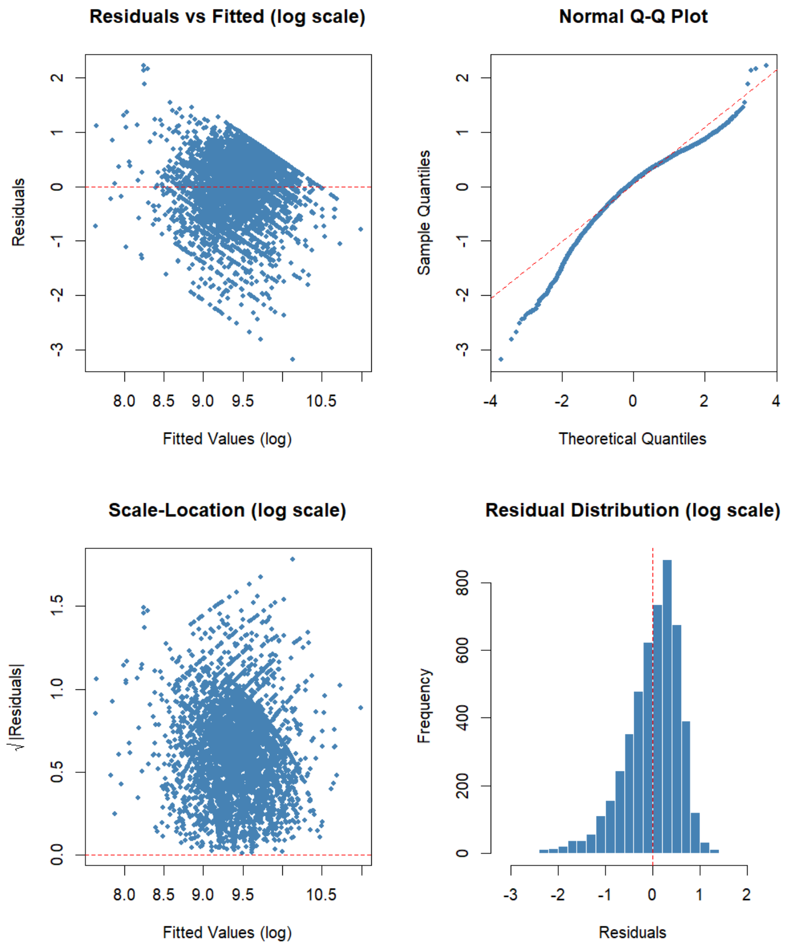
Biggest Limitation: Model Results!

RMSE: 7663.22 **MAPE: 55.73%** **R2: 0.25**

The model performs average at best but can only explain 25% of the variance in the data.

Either, the relationships are more complex and non-linear, or business context matters just as much as credit history

Project Review (Data Problems)



The non-normal distribution of the target variable (a key assumption of a linear regression) caused cascading issues in the model

Large clusters meant it was less of a continuous distribution and more of a discrete distribution

The relationships to the target variable were not very linear, log transformations helped improve result, but didn't solve the issue

Impact

The deployment of a model like this has the power to

- Increase lending accuracy
- Remove human biases and increase equity
- Reduce the risk to GE

However, with the poor performance of the current version it could

- Decrease lending accuracy
- Increase noise and bias
- Increase the risk to GE

With GE being an international company, with upwards of tens of thousands of customers, having an average error of $7,500 means putting $75,000,00 of error at risk, not including the lost customers and defaults on credit

Recommendations

Two new approaches to build upon this project

Increase the breadth of data being considered in this model

- Include business context from the customer on what the money is being used for
- Include the amount of money the customer asks for

Change to a different kind of model

- Assume that the distribution is discrete and/or non-normal
- Utilize ensembles to capture more complex relationships

Recommendations

Had the model proven better results, this project could have alternative uses

- It could be used to gauge risk (if a customer asks for more money than the model predicts they may be more risky)
- It could detect loan amount that are exceedingly large and flag it for further review (Fraud detection)
- Using the weights of the model would help management determine what factors drive the dollar amount of a loan (inference)